

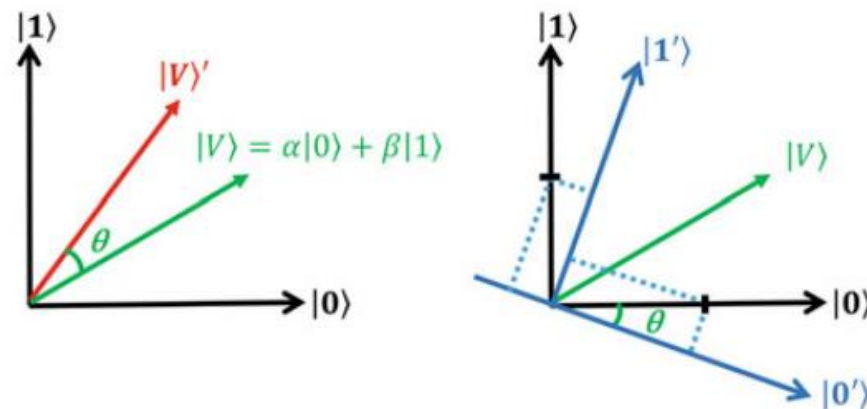
Chapter 9:

Unitary Transformation, Completeness, and Construction of Operator

Kamonluk Suksen Ph.D.

Unitary Transformation

- The definition of a unitary matrix, U , is that it satisfies $U^\dagger U = I$, when two vectors are transformed by the same unitary their inner product is unchanged. It means that all the vectors (states) are transformed by the same unitary vector, it is equivalent to transforming the coordinates in the opposite “direction”.
- If we rotate all vectors in the same way, it is equivalent to rotating the coordinates or the hyperspace in the opposite way. This is just like if there is only the Earth in an empty space, you can say the plane is flying around a fixed Earth or the Earth is rotating, in the opposite direction, under a stationary plane.
- Naturally, a transformation by a unitary matrix is called a **unitary transformation**.
- Then it is also the most important when we perform coordinates (basis) transformation, but in the opposite direction. Whenever we talk about unitary transformation, we apply it to coordinate transformation.



Pic.7 Vector $|V\rangle$ rotated anti-clockwise by θ (left) is equivalent to the basis vectors $|0\rangle / |1\rangle$ rotated clockwise by θ

Unitary Transformation

- We know any vector $|V\rangle$ can be represented as a linear combination (i.e. superposition) of the basis states and we write it as:

$$|V\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

- Just remind that we are borrowing the 2D plane for visualization and $|0\rangle$ and $|1\rangle$ are orthogonal to each other and they cannot co-exist with 100% certainty at the same time.
- When we perform a measurement, $|V\rangle$ will collapse to either $|0\rangle$ or $|1\rangle$ with probabilities $|\alpha|^2$ and $|\beta|^2$, respectively.
- If we rotate (transform) $|V\rangle$ anti-clockwise by θ (Left of Pic. 7), it becomes $|V'\rangle$. How do we represent it in the old basis in terms of the old basis states $|0\rangle$ and $|1\rangle$?
- By using trigonometry, we can find the new representation. But for another view. Instead of rotating $|V\rangle$ anti-clockwise by θ , we can also rotate the basis (i.e. its basis states $|0\rangle$ and $|1\rangle$) clockwise by θ (Right of Pic. 7). The representation of $|V\rangle$ by the new basis states $|0'\rangle$ and $|1'\rangle$ will then be equal to the representation of $|V'\rangle$ by the old basis states $|0\rangle$ and $|1\rangle$.
- The coordinate transformation is just another name for the basis change.

Unitary Transformation

- From the right of Pic. 7, using trigonometry,

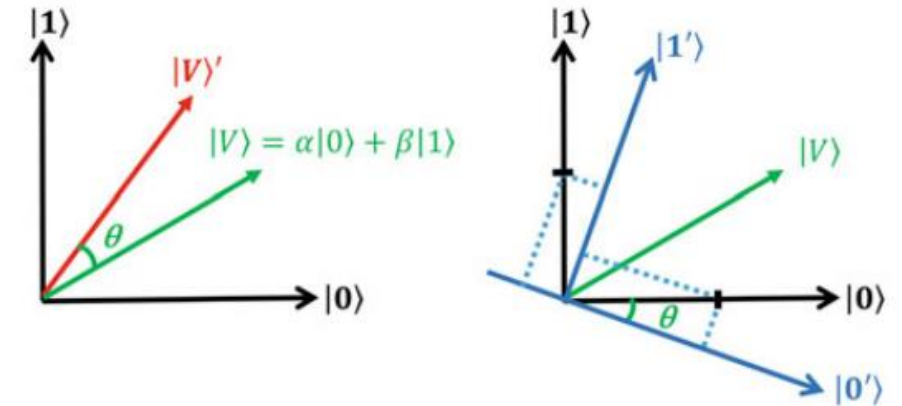
$$|0\rangle = \cos \theta |0'\rangle + \sin \theta |1'\rangle = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

$$|1\rangle = -\sin \theta |0'\rangle + \cos \theta |1'\rangle = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$$

The basis we are using now is the $|0'\rangle$ and $|1'\rangle$ basis.

$$\begin{aligned} |V\rangle &= \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \\ &= \alpha(\cos \theta |0'\rangle + \sin \theta |1'\rangle) + \beta(-\sin \theta |0'\rangle + \cos \theta |1'\rangle) \\ &= (\alpha \cos \theta - \beta \sin \theta) |0'\rangle + (\alpha \sin \theta + \beta \cos \theta) |1'\rangle \\ &= \begin{pmatrix} \alpha \cos \theta - \beta \sin \theta \\ \alpha \sin \theta + \beta \cos \theta \end{pmatrix}_{|0'\rangle/|1'\rangle} \end{aligned}$$

- For the same $|V\rangle$, it can have two different column forms because it is represented in two different bases. In this process, we have found a matrix to transform the representation in the old basis to the new basis.



Pic.7 Vector $|V\rangle$ rotated anti-clockwise by θ (left) is equivalent to the basis vectors $|0\rangle / |1\rangle$ rotated clockwise by θ

Unitary Transformation

- For the same $|V\rangle$, it can have two different column forms because it is represented in two different bases. In this process, we have found a matrix to transform the representation in the old basis to the new basis,

$$\begin{pmatrix} \alpha \cos \theta - \beta \sin \theta \\ \alpha \sin \theta + \beta \cos \theta \end{pmatrix}_{|0'\rangle/|1'\rangle} = \mathbf{U} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}_{|0\rangle/|1\rangle}$$

where

$$\mathbf{U} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Construction of Unitary Transformation Matrices

- How do we construct the transformation matrix?
- Let the basis vectors in the old basis be $|0\rangle, |1\rangle, \dots, |n-1\rangle$ and those in the new basis be $|0'\rangle, |1'\rangle, \dots, |n-1'\rangle$ in an n -dimensional space. The transformation matrix to represent a vector in the new basis is given by

$$U = \begin{pmatrix} \langle 0'|0\rangle & \langle 0'|1\rangle & \cdots & \langle 0'|n-1\rangle \\ \langle 1'|0\rangle & \langle 1'|1\rangle & \cdots & \langle 1'|n-1\rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle n-1'|0\rangle & \langle n-1'|1\rangle & \cdots & \langle n-1'|n-1\rangle \end{pmatrix}$$

The i -th row j -th column element is just the inner product of the i -th new basis vector and the j -th old basis vector, $\langle i'|j\rangle$. For example:

$$\begin{aligned} \langle 0'|0\rangle &= \langle 0'|(\cos \theta |0'\rangle + \sin \theta |1'\rangle) = \cos \theta \\ \langle 0'|1\rangle &= \langle 0'|(-\sin \theta |0'\rangle + \cos \theta |1'\rangle) = -\sin \theta \\ \langle 1'|0\rangle &= \langle 1'|(\cos \theta |0'\rangle + \sin \theta |1'\rangle) = \sin \theta \\ \langle 1'|1\rangle &= \langle 1'|(-\sin \theta |0'\rangle + \cos \theta |1'\rangle) = \cos \theta \end{aligned}$$

We know that $|0'\rangle, |1'\rangle$ are orthonormal and used the distributive property. Therefore,

$$U = \begin{pmatrix} \langle 0'|0\rangle & \langle 0'|1\rangle \\ \langle 1'|0\rangle & \langle 1'|1\rangle \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Completeness of Basis

- In quantum computing, the basis needs to be orthonormal. It makes our life much easier when deriving equations (e.g. it allows us to use $\langle 1|0\rangle = 0$). Another important requirement for the basis is its **completeness**.
- This just means that the basis should span the whole space we are interested in.
- For example, a basis composed of $|\hat{x}\rangle$ and $|\hat{y}\rangle$ is complete for a real 2D space but incomplete for a real 3D space.
- Every vector in the real 2D space can be represented as a linear combination of $|\hat{x}\rangle$ and $|\hat{y}\rangle$, but some vectors in the real 3D space cannot. When the basis is complete, we can decompose any vector in the space of interest into a linear combination of the basis vectors.
- For example, in an n -D space, a vector $|v\rangle$ can be described as the linear combination of a basis composed of basis vectors $|i\rangle$, where i runs from 0 to $n - 1$.

$$|v\rangle = \sum_{i=0}^{n-1} \alpha_i |i\rangle$$

where α_i are scalars (complex numbers) and describe “how much” $|i\rangle$ vector $|v\rangle$ has. How to find α_i (which is one of the α_i inside the summation)?

Completeness of Basis

- With Dirac's bra-ket notation, this is very straightforward and we just need to perform the inner product of $|j\rangle$ and $|i\rangle$ and take advantage of the orthonormal property of the basis vectors,

$$\langle j|V\rangle = \sum_{i=0}^{n-1} \alpha_i \langle j|i\rangle = \sum_{i=0}^{n-1} \alpha_i \delta_{ji} = \alpha_j$$

- Or we can project $|V\rangle$ on to $|j\rangle$ to find the resulting vector. As we know that $P_j = |j\rangle\langle j|$. Therefore,

$$\begin{aligned} P_j|V\rangle &= |j\rangle\langle j||V\rangle \\ &= \sum_{i=0}^{n-1} \alpha_i |j\rangle\langle j|i\rangle = \sum_{i=0}^{n-1} \alpha_i |j\rangle \delta_{ji} = \alpha_j |j\rangle \end{aligned}$$

Note $\langle j|i\rangle$ is a scalar and we can move it freely to the front inside the summation. It gives the same result as the first equation except it gives the projected vector instead of just the “amount” of the component $|j\rangle$.

Completeness of Basis

- From $\langle j|V\rangle = \alpha_j$, we can substitute it in $|v\rangle = \sum_{i=0}^{n-1} \alpha_i |i\rangle$, but rename j to i :

$$\begin{aligned}
 |v\rangle &= \sum_{i=0}^{n-1} \alpha_i |i\rangle \\
 &= \sum_{i=0}^{n-1} \langle i|V\rangle |i\rangle \\
 &= \sum_{i=0}^{n-1} |i\rangle \langle i|V\rangle \\
 &= \left(\sum_{i=0}^{n-1} |i\rangle \langle i| \right) |V\rangle
 \end{aligned}$$

Since it gives $|v\rangle$ when it is multiplied to an arbitrary vector $|v\rangle$, it means it must be the identity matrix.

$$\sum_{i=0}^{n-1} |i\rangle \langle i| = I$$

- It tells us that the outer products of each basis to itself sum up to an identity matrix.

Construct Operator from Eigenvalues and Eigenvectors

- If an operator has eigenvalues λ_i and eigenvectors $|i\rangle$, where i runs from 0 to $n - 1$, the operator, A , is given by,

$$A = \sum_{i=0}^{n-1} \lambda_i |i\rangle\langle i|$$

- When the matrix is represented in the basis formed by its eigenvectors, this is just the diagonal matrix.
- Let us consider σ_z . Its eigenvectors are $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ with eigenvalues of 1 and -1, respectively. Then it means

$$\begin{aligned} \sigma_z &= \sum_{i=0}^{2-1} \lambda_i |i\rangle\langle i| \\ &= \lambda_0 |0\rangle\langle 0| + \lambda_1 |1\rangle\langle 1| \\ &= 1 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} (1 \ 0) + -1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} (0 \ 1) \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

This is exactly the matrix form of σ_z .

Summary

- **Representation in Different Bases:** The core concept emphasized is representing quantum states in various bases, analogous to expressing the exact same meaning in different languages (e.g., English vs. Cantonese).
- **Role of Unitary Matrices:** Unitary matrices are used to transform vectors from one representation to another by rotating the basis in hyperspace.
- **The Translation Analogy:** In this context, the matrix acts as a "dictionary," and the rotation represents the actual transformation or translation process between the two "languages" (bases).
- **Mathematical Foundations:** The text notes the importance of the concept of completeness and the ability to construct matrices using eigenvalues and eigenvectors.
- **Relevance to Quantum Computing:** These mathematical skills are fundamental tools that will be essential for future quantum computing applications.

Chapter 10: Hilbert Space, Tensor Product, and Multi-Qubit

Kamonluk Suksen Ph.D.

Hilbert Space

- A **Hilbert space** is a complete linear space with an inner product defined. The word “complete” here means it converges in certain operations and it is not the same as the completeness concept.
- It is just a generalization of our real 3D space. So we can loosely regard a Hilbert space as a space with any dimensions which allows complex coefficients.
- In this space, it has vectors and it has a well-defined inner product, $\langle f|g\rangle$, of any pair of vectors, $|f\rangle$ and $|g\rangle$, and for any complex number, c , with the following properties

$$\langle f|f\rangle \geq 0$$

The inner product of a vector to itself has to be positive or zero. This is because $\langle f|f\rangle$ is just the square of the length of the vector $|f\rangle$.

- We can scale $|f\rangle$ by a scalar c and then take its bra version to multiply to $|g\rangle$, it is the same as multiplying $|f\rangle$'s bra version on $|g\rangle$ followed by multiplying the complex conjugate of c . So, we get

$$\langle cf|g\rangle = c^* \langle f|g\rangle$$

But no complex conjugate is needed if the c is used to scale the second vector, which only appears in its ket version as following:

$$\langle f|cg\rangle = c \langle f|g\rangle$$

- As we know that projecting $|g\rangle$ on $|f\rangle$ gives a scalar which is the complex conjugate of the projection of $|f\rangle$ on $|g\rangle$. And the last equation is just the distribution properties of a linear operation.

$$\langle f|g\rangle = \langle g|f\rangle^*$$

$$\langle f + h|g\rangle = \langle f|g\rangle + \langle h|g\rangle$$

Expansion of Hilbert Space and Tensor Product

- If there is one electron, we know that its spin forms a Hilbert space with the basis vectors of spinning up or down (i.e. $|0\rangle$ or $|1\rangle$). We say it is a Hilbert space because it fulfills the complete requirement and the requirements from previous equations. Every vector in this space is just a linear combination of the basis states, $\alpha|0\rangle + \beta|1\rangle$.
- We may also use $|+\rangle/|-\rangle$ as the basis but it still the same Hilbert space (just different representations). Let's call this space, $\mathcal{H}_1$.
- How to describe the position of the electron? The position itself also forms a Hilbert space with the basis vectors being any point in the real space. The state of the electron in this space is a superposition of the basis vectors.
- In principle, for example, the electron in a water molecule in the cloud has a finite probability to be in your cup as $|Location\ state\ of\ electron\rangle = \alpha|At\ a\ point\ in\ the\ cloud\rangle + \beta|At\ a\ point\ in\ your\ cup\rangle + \dots$, although β can be very small. Let's call this space, $\mathcal{H}_2$.
- The spin Hilbert space and the position Hilbert space can form a bigger (higher-dimensional) space, $\mathcal{H}_3$. The way to form it is by using the tensor product denoted as $\otimes$,

$$\mathcal{H}_3 = \mathcal{H}_1 \otimes \mathcal{H}_2$$

Expansion of Hilbert Space and Tensor Product

- We also use the tensor product of two smaller spaces to form the vector in the larger space.
- If $|f\rangle_1$ and $|g\rangle_2$ are two vectors in $\mathcal{H}_1$ and $\mathcal{H}_2$, respectively, it can form a vector $|h\rangle_3$ in $\mathcal{H}_3$ through $|h\rangle_3 = |f\rangle_1 \otimes |g\rangle_2$.
- The tensor product is what makes quantum computing so powerful. Note that the spin space has *no* interaction with the position space. We can treat them as two quantities in two different universes. They never talk to each other. So, the basis vectors in the spin space and the position space can form a new basis in $\mathcal{H}_3$ through permutation.
- Assume we have two electrons. The first electron has its own spin space with basis vectors $|0\rangle_1$ and $|1\rangle_1$. The subscript “1” to denote that the basis vectors are of the first electron or in Hilbert space $\mathcal{H}_1$. The state of the first electron can be a linear combination of the basis states,

$$|f\rangle_1 = \alpha_1|0\rangle_1 + \beta_1|1\rangle_1$$

- Similarly, for the second electron, it has its own spin space $\mathcal{H}_2$ and basis vectors $|0\rangle_2$ and $|1\rangle_2$ and any spin states of the second electron $|g\rangle_2$ can be represented as

$$|g\rangle_2 = \alpha_2|0\rangle_2 + \beta_2|1\rangle_2$$

Expansion of Hilbert Space and Tensor Product

- To describe the system of the two electrons which have individual spins, we can use a tensor product, $\mathcal{H}_3 = \mathcal{H}_1 \otimes \mathcal{H}_2$. What are the basis states of this new system with two electrons? It will be the permutation through product of their basis states,

$$|0\rangle_3 = |0\rangle_1 \otimes |0\rangle_2 = |0\rangle_1 |0\rangle_2$$

$$|1\rangle_3 = |0\rangle_1 \otimes |1\rangle_2 = |0\rangle_1 |1\rangle_2$$

$$|2\rangle_3 = |1\rangle_1 \otimes |0\rangle_2 = |1\rangle_1 |0\rangle_2$$

$$|3\rangle_3 = |1\rangle_1 \otimes |1\rangle_2 = |1\rangle_1 |1\rangle_2$$

- $\mathcal{H}_3$ has four basis states and they are just the permutation of electron 1's and electron 2's basis states.
- How to represent an arbitrary state $|h\rangle_3$ in $\mathcal{H}_3$? Again it must be a linear combination of the basis vectors in $\mathcal{H}_3$. Therefore,

$$|h\rangle_3 = \alpha_3 |0\rangle_3 + \beta_3 |1\rangle_3 + \gamma_3 |2\rangle_3 + \delta_3 |3\rangle_3$$

where α_3 , β_3 , γ_3 , and δ_3 are the complex coefficients.

Expansion of Hilbert Space and Tensor Product

- By substituting, we can write state $|h\rangle_3$ in $\mathcal{H}_3$ as:

$$\begin{aligned}
 |h\rangle_3 &= \alpha_3|0\rangle_1 \otimes |0\rangle_2 + \beta_3|0\rangle_1 \otimes |1\rangle_2 + \gamma_3|1\rangle_1 \otimes |0\rangle_2 + \delta_3|1\rangle_1 \otimes |1\rangle_2 \\
 &= \alpha_3|0\rangle_1|0\rangle_2 + \beta_3|0\rangle_1|1\rangle_2 + \gamma_3|1\rangle_1|0\rangle_2 + \delta_3|1\rangle_1|1\rangle_2 \\
 &= \alpha_3|0\rangle|0\rangle + \beta_3|0\rangle|1\rangle + \gamma_3|1\rangle|0\rangle + \delta_3|1\rangle|1\rangle \\
 &= \alpha_3|00\rangle + \beta_3|01\rangle + \gamma_3|10\rangle + \delta_3|11\rangle
 \end{aligned}$$

- We can also write it in column form,

$$|h\rangle_3 = \begin{pmatrix} \alpha_3 \\ \beta_3 \\ \gamma_3 \\ \delta_3 \end{pmatrix}$$

Multi-Qubits

- Qubit stands for *quantum bit*. It refers to a vector space with two basis vectors and it is 2D. We are very familiar with this already. The vector in this vector space can be represented by a qubit, which is the linear combination of the basis vectors. And the basis vectors can be $|0\rangle/|1\rangle$ and $|+\rangle/|-\rangle$ in a single spin system.
- We just discussed that we can form a larger system with two qubits and the number of basis vectors becomes four. Since the tensor product is used to construct a larger system from its subsystems through the permutation of basis vectors, the number of basis states grows exponentially and this is the power of quantum computing.
- For a single qubit, we call it a $\mathbb{C}^2$ space, which means it is a 2D complex vector space ($\mathbb{C}$ means complex). When there are 2 qubits, the space is $\mathbb{C}^4$ as it is a 4-dimensional due to the tensor product of two $\mathbb{C}^2$ spaces. E.g. for n qubits, it is a tensor product of two $n\mathbb{C}^2$ spaces and the dimension is 2^n instead of n . So, if there are 10 qubits, the dimension is now 1024 and there are 1024 basis states! A vector (state) in the 10-qubit system will be a superposition of 1024 basis states which will enable **quantum parallelism**.
- Since it is just the permutation of the basis vectors of single-qubits, then it is just the permutation of 0's and 1's. Therefore, they are just the binary numbers from 0 to $2^n - 1$ for n -qubits. This helps us to memorize the basis vectors and also allows us to write them succinctly in decimal numbers in the kets.
- Moreover, we need to be very careful how each qubit is ordered. Which qubit is treated as the Most Significant Bit (MSB) and which is the Least Significant Bit (LSB). For the spins of two electrons, A and B , we may write electron A on the left (as the MSB) or on the right (as the LSB). For MSB case, the tensor product state due to electron A spinning up and electron B spinning down is $|0\rangle_A|1\rangle_B = |01\rangle = |1\rangle$. For LSB case, is $|1\rangle_B|0\rangle_A = |10\rangle = |2\rangle$

Multi-Qubits

- Let's understand vector tensor product from another perspective, i.e. in their column forms.

$$|h\rangle_3 = |f\rangle_1 \otimes |g\rangle_2$$

$$\begin{pmatrix} \alpha_3 \\ \beta_3 \\ \gamma_3 \\ \delta_3 \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \otimes \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix}$$

The column vector form clearly shows that how two $\mathbb{C}^2$ can form a $\mathbb{C}^4$ vector. But how do we find the $\begin{pmatrix} \alpha_3 \\ \beta_3 \\ \gamma_3 \\ \delta_3 \end{pmatrix}$ for the given $\begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix}$ and $\begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix}$?

$$\begin{aligned} |h\rangle_3 &= \alpha_3|0\rangle_1 \otimes |0\rangle_2 + \beta_3|0\rangle_1 \otimes |1\rangle_2 + \gamma_3|1\rangle_1 \otimes |0\rangle_2 + \delta_3|1\rangle_1 \otimes |1\rangle_2 \\ &= \alpha_3|0\rangle_1|0\rangle_2 + \beta_3|0\rangle_1|1\rangle_2 + \gamma_3|1\rangle_1|0\rangle_2 + \delta_3|1\rangle_1|1\rangle_2 \\ &= |f\rangle_1 \otimes |g\rangle_2 \\ &= (\alpha_1|0\rangle_1 + \beta_1|1\rangle_1)(\alpha_2|0\rangle_2 + \beta_2|1\rangle_2) \\ &= \alpha_1\alpha_2|0\rangle_1|0\rangle_2 + \alpha_1\beta_2|0\rangle_1|1\rangle_2 + \beta_1\alpha_2|1\rangle_1|0\rangle_2 + \beta_1\beta_2|1\rangle_1|1\rangle_2 \end{aligned}$$

- So, $\alpha_3 = \alpha_1\alpha_2$; $\beta_3 = \alpha_1\beta_2$; $\gamma_3 = \beta_1\alpha_2$; and $\delta_3 = \beta_1\beta_2$.

Multi-Qubits

- Therefore, this is just a permutation of the coefficients through scalar multiplication. So, we can also do this in column vector using the following steps

$$\begin{pmatrix} \alpha_3 \\ \beta_3 \\ \gamma_3 \\ \delta_3 \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \otimes \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix}$$

$$= \begin{pmatrix} \alpha_1 \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} \\ \beta_1 \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \alpha_1 \alpha_2 \\ \alpha_1 \beta_2 \\ \beta_1 \alpha_2 \\ \beta_1 \beta_2 \end{pmatrix}$$

Multi-Qubits

- Basis vectors of multi-qubit systems

Number of qubits	Number of basis vectors	Basis Vectors	Space
1	2	$ 0\rangle, 1\rangle$	$\mathbb{C}^2$
2	4	$ 0\rangle 0\rangle, 0\rangle 1\rangle, 1\rangle 0\rangle, 1\rangle 1\rangle$ Or: $ 00\rangle, 01\rangle, 10\rangle, 11\rangle$ Or: $ 0\rangle, 1\rangle, 2\rangle, 3\rangle$	$\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$
3	8	$ 0\rangle 0\rangle 0\rangle, 0\rangle 0\rangle 1\rangle, \dots, 1\rangle 1\rangle 1\rangle$ Or: $ 000\rangle, 001\rangle, \dots, 111\rangle$ Or: $ 0\rangle, 1\rangle, \dots, 7\rangle$	$\mathbb{C}^8 = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$
n	2^n	$ 0\rangle, 1\rangle, \dots, 2^n - 1\rangle$	$\mathbb{C}^{2^n} = \mathbb{C}^2 \otimes \dots \otimes \mathbb{C}^2 = (\mathbb{C}^2)^{n \otimes}$

More About Tensor Product in Hilbert Space

- If $\mathcal{H}_3 = \mathcal{H}_1 \otimes \mathcal{H}_2$, what vectors does $\mathcal{H}_3$ contains? It contains the linear combination of the tensor products of the vectors in $\mathcal{H}_1$ and $\mathcal{H}_2$.

$$|h\rangle = \sum_{j=1}^n |f_i\rangle \otimes |g_j\rangle$$

- For example, n can be 1 and $|h\rangle$ can be $|0\rangle |1\rangle$ or $|1\rangle |1\rangle$. Or n can be 2, then $|h\rangle$ may be $i |0\rangle |1\rangle + |1\rangle |0\rangle$, if $|f_1\rangle = i|0\rangle$, $|f_2\rangle = |1\rangle$, $|g_1\rangle = |1\rangle$, and $|g_2\rangle = |0\rangle$.
- We may also scale the vector in their subspace first and then perform tensor product or vice versa. The result is the same,

$$c|f\rangle \otimes |g\rangle = |cf\rangle \otimes |g\rangle = |f\rangle \otimes |cg\rangle$$

It also obeys distribution rule, from both right and left,

$$(|f\rangle + |e\rangle) \otimes |g\rangle = |f\rangle \otimes |g\rangle + |e\rangle \otimes |g\rangle$$

$$|g\rangle \otimes (|f\rangle + |e\rangle) = |g\rangle \otimes |f\rangle + |g\rangle \otimes |e\rangle$$

More About Tensor Product in Hilbert Space

- Finally, if we perform inner product in their respective $\mathcal{H}_1$ and $\mathcal{H}_2$ spaces first and then perform the tensor product, it is the same as forming tensor product first and then perform inner product in the $\mathcal{H}_3$ space.

$$\langle h_1 | h_2 \rangle = \left\langle \sum_{j=1}^{n_1} |f_j\rangle \otimes |g_j\rangle \middle| \sum_{l=1}^{n_2} |e_l\rangle \otimes |k_l\rangle \right\rangle = \sum_{j=1}^{n_1} \sum_{l=1}^{n_2} \langle f_j | e_l \rangle \langle g_j | k_l \rangle$$

Here, $|h_1\rangle$ and $|h_2\rangle$ are the vectors in the $\mathcal{H}_3$ space. They are formed by the linear combination of tensor products of $|f_j\rangle$ and $|e_l\rangle$ in the $\mathcal{H}_1$ space and $|g_j\rangle$ and $|k_l\rangle$ in the $\mathcal{H}_2$. This shows again the subspaces are not “talking to each other” because $|f_j\rangle$ and $|g_j\rangle$ only form inner products with $|e_l\rangle$ and $|k_l\rangle$, respectively.

Summary

- **Hilbert Space Fundamentals**
 - Many properties are similar to the familiar 3D space we live in.
 - Operations overlap with concepts learned in previous chapters; memorizing every property is not necessary.
- **The Power of Tensor Product**
 - Enables the construction of higher-dimensional Hilbert spaces.
 - Allows us to produce a vector (state) in a larger space using given lower-dimensional vectors.
- **Why This Matters**
 - These concepts serve as the essential foundation for understanding Quantum Computing Algorithms.

Chapter 11:

Tensor Product of Operators, Partial Measurement, and Matrix Representation in a Given Basis

Kamonluk Suksen Ph.D.

Tensor Product of Vectors in General Form

- How to perform tensor product for two vectors from arbitrary/higher-dimensional spaces?
- Let $\vec{V} = V_0\hat{e}_0 + V_1\hat{e}_1 + \cdots + V_{n-1}\widehat{e_{n-1}}$ in an n -dimensional space and $\vec{W} = W_0\hat{f}_0 + W_1\hat{f}_1 + \cdots + W_{m-1}\widehat{f_{m-1}}$ in an m -dimensional space. $\hat{e}_0, \dots, \widehat{e_{n-1}}$ and $\hat{f}_0, \dots, \widehat{f_{m-1}}$ are the basis vectors in the corresponding space.
- We can write it in bra-ket notation or column form, too.

$$|\vec{V}\rangle = V_0|\hat{e}_0\rangle + V_1|\hat{e}_1\rangle + \cdots + V_{n-1}|\widehat{e_{n-1}}\rangle = \begin{pmatrix} V_0 \\ V_1 \\ \vdots \\ V_{n-1} \end{pmatrix}$$

$$|\vec{W}\rangle = W_0|\hat{f}_0\rangle + W_1|\hat{f}_1\rangle + \cdots + W_{m-1}|\widehat{f_{m-1}}\rangle = \begin{pmatrix} W_0 \\ W_1 \\ \vdots \\ W_{m-1} \end{pmatrix}$$

- If we perform a tensor product of $|\vec{V}\rangle \otimes |\vec{W}\rangle$, what do we get?

$$\begin{aligned} |\vec{V}\rangle \otimes |\vec{W}\rangle &= (V_0|\hat{e}_0\rangle + V_1|\hat{e}_1\rangle + \cdots + V_{n-1}|\widehat{e_{n-1}}\rangle) \otimes (W_0|\hat{f}_0\rangle + W_1|\hat{f}_1\rangle + \cdots + W_{m-1}|\widehat{f_{m-1}}\rangle) \\ &= V_0W_0|\hat{e}_0\rangle|\hat{f}_0\rangle + V_0W_1|\hat{e}_0\rangle|\hat{f}_1\rangle + \cdots + V_1W_0|\hat{e}_1\rangle|\hat{f}_0\rangle + \cdots + V_{n-1}W_{m-1}|\widehat{e_{n-1}}\rangle|\widehat{f_{m-1}}\rangle \end{aligned}$$

Tensor Product of Vectors in General Form

- So how many new basis vectors do we have now? There are $|\widehat{e}_0\rangle|\widehat{f}_0\rangle, |\widehat{e}_0\rangle|\widehat{f}_1\rangle, \dots, |\widehat{e}_1\rangle|\widehat{f}_0\rangle, \dots, |\widehat{e}_{n-1}\rangle|\widehat{f}_{m-1}\rangle$.
- They are the permutation of the basis vectors from the two spaces and there are $n \times m$ new basis vectors. So, the new space is $n \times m$ –dimensional. We can see this from the column vector multiplication in a tensor product.

$$\begin{aligned}
 |\vec{V}\rangle \otimes |\vec{W}\rangle &= \begin{pmatrix} V_0 \\ V_1 \\ \vdots \\ V_{n-1} \end{pmatrix} \otimes \begin{pmatrix} W_0 \\ W_1 \\ \vdots \\ W_{m-1} \end{pmatrix} \\
 &= \begin{pmatrix} V_0 \begin{pmatrix} W_0 \\ W_1 \\ \vdots \\ W_{m-1} \end{pmatrix} \\ \vdots \\ V_{n-1} \begin{pmatrix} W_0 \\ W_1 \\ \vdots \\ W_{m-1} \end{pmatrix} \end{pmatrix} = \begin{pmatrix} V_0 W_0 \\ \vdots \\ V_0 W_{m-1} \\ \vdots \\ V_{n-1} W_0 \\ \vdots \\ V_{n-1} W_{m-1} \end{pmatrix}
 \end{aligned}$$

Tensor Product of Vectors in General Form

- Sometimes the tensor product is written in matrix form to make it more compact.

$$|\vec{V}\rangle \otimes |\vec{W}\rangle = \begin{pmatrix} V_0 \\ V_1 \\ \vdots \\ V_{n-1} \end{pmatrix} (W_0 \dots \dots W_{m-1}) = \begin{pmatrix} V_0 W_0 & \dots & V_0 W_{m-1} \\ \vdots & \ddots & \vdots \\ V_{n-1} W_0 & \dots & V_{n-1} W_{m-1} \end{pmatrix}$$

In this type of notation, the second vector is written in row form. And the basis vector for the i -th row and j -th column element in the matrix is $|\hat{e}_i\rangle |\hat{f}_j\rangle$

Tensor Product of Operators

- In each Hilbert space, it has its own operators. Physically, if there are two electrons, there will be two separate operators used to manipulate the spin of the electrons (electrons to rotate the vector in the $\mathbb{C}^2$ space).
- We use a magnetic field to manipulate the spin of the electron. So we need two magnetic fields to manipulate the spin of two electrons.
- In the $\mathbb{C}^2$ spaces, it just means that there are two matrices to rotate the vectors, one in each space and each corresponds to a different set of basis vectors.
- If we consider the two electron as a whole system, we can use the tensor product to construct the state in the larger system. How do we construct an operator to describe the manipulation of the combined vector? That is how to describe the two magnetic fields used to control the two electrons as a whole. We still use the tensor product.
- Assume $\mathcal{H}_3 = \mathcal{H}_1 \otimes \mathcal{H}_2$, and $\mathbf{U}_1$ and $\mathbf{U}_2$ are two operators in the two $\mathbb{C}^2$ spaces of $\mathcal{H}_1$ and $\mathcal{H}_2$, respectively. Assume,

$$\mathbf{U}_1 = \begin{pmatrix} a & c \\ b & d \end{pmatrix}; \quad \mathbf{U}_2 = \begin{pmatrix} e & g \\ f & h \end{pmatrix}$$

Tensor Product of Operators

- The corresponding operator in the $\mathcal{H}_3$ space is $U = U_1 \otimes U_2$, where

$$\begin{aligned}
 U = U_1 \otimes U_2 &= \begin{pmatrix} a & c \\ b & d \end{pmatrix} \otimes \begin{pmatrix} e & g \\ f & h \end{pmatrix} \\
 &= \begin{pmatrix} a \begin{pmatrix} e & g \\ f & h \end{pmatrix} & c \begin{pmatrix} e & g \\ f & h \end{pmatrix} \\ b \begin{pmatrix} e & g \\ f & h \end{pmatrix} & d \begin{pmatrix} e & g \\ f & h \end{pmatrix} \end{pmatrix} = \begin{pmatrix} ae & ag & ce & cg \\ af & ah & cf & ch \\ be & bg & de & dg \\ bf & bh & df & dh \end{pmatrix} \begin{matrix} |00'\rangle \\ |01'\rangle \\ |10'\rangle \\ |11'\rangle \end{matrix}
 \end{aligned}$$

We also labeled the basis states next to the matrix so we know which basis vector each row corresponds to.

- For example, assume both electrons have an up-spin, i.e. $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|0'\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. If we consider each electron individually and apply operators on the vectors in the individual $\mathbb{C}^2$ spaces,

$$\begin{aligned}
 U_1|0\rangle &= \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} = a|0\rangle + b|1\rangle \\
 U_2|0'\rangle &= \begin{pmatrix} e & g \\ f & h \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} e \\ f \end{pmatrix} = e|0'\rangle + f|1'\rangle
 \end{aligned}$$

Tensor Product of Operators

- We can then form a tensor product to see how the final vector looks like in the $\mathbb{C}^4$ spaces,

$$\begin{aligned} U_1|0\rangle \otimes U_2|0'\rangle &= (a|0\rangle + b|1\rangle) \otimes (e|0'\rangle + f|1'\rangle) \\ &= \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} e \\ f \end{pmatrix} = \begin{pmatrix} a \begin{pmatrix} e \\ f \end{pmatrix} \\ b \begin{pmatrix} e \\ f \end{pmatrix} \end{pmatrix} = \begin{pmatrix} ae \\ af \\ be \\ bf \end{pmatrix} \begin{matrix} |00'\rangle \\ |01'\rangle \\ |10'\rangle \\ |11'\rangle \end{matrix} \end{aligned}$$

- Or we can also operate e directly on the tensor product of $|0\rangle$ and $|0'\rangle$ (this initial states). We get $|00'\rangle$, and then apply U to $|00'\rangle$, i.e. to apply an operator on a vector in the $\mathbb{C}^4$ spaces.

$$U|00'\rangle = \begin{pmatrix} ae & ag & ce & cg \\ af & ah & cf & ch \\ be & bg & de & dg \\ bf & bh & df & dh \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} ae \\ af \\ be \\ bf \end{pmatrix} \begin{matrix} |00'\rangle \\ |01'\rangle \\ |10'\rangle \\ |11'\rangle \end{matrix}$$

The result is exactly the same as previous.

Tensor Product of Operators

- In general, the tensor product of an n -dimensional operator and a m -dimensional
- operator is an $n \times m$ -dimensional operator.

$$\begin{aligned}
 & \begin{pmatrix} a_{0,0} & \cdots & a_{0,n-1} \\ \vdots & \ddots & \vdots \\ a_{n-1,0} & \cdots & a_{n-1,n-1} \end{pmatrix} \otimes \begin{pmatrix} b_{0,0} & \cdots & b_{0,m-1} \\ \vdots & \ddots & \vdots \\ b_{m-1,0} & \cdots & b_{m-1,m-1} \end{pmatrix} \\
 &= \begin{pmatrix} a_{0,0} \begin{pmatrix} b_{0,0} & \cdots & b_{0,m-1} \\ \vdots & \ddots & \vdots \\ b_{m-1,0} & \cdots & b_{m-1,m-1} \end{pmatrix} & \cdots & a_{0,n-1} \begin{pmatrix} b_{0,0} & \cdots & b_{0,m-1} \\ \vdots & \ddots & \vdots \\ b_{m-1,0} & \cdots & b_{m-1,m-1} \end{pmatrix} \\ \vdots & \ddots & \vdots \\ a_{n-1,0} \begin{pmatrix} b_{0,0} & \cdots & b_{0,m-1} \\ \vdots & \ddots & \vdots \\ b_{m-1,0} & \cdots & b_{m-1,m-1} \end{pmatrix} & \cdots & a_{n-1,n-1} \begin{pmatrix} b_{0,0} & \cdots & b_{0,m-1} \\ \vdots & \ddots & \vdots \\ b_{m-1,0} & \cdots & b_{m-1,m-1} \end{pmatrix} \end{pmatrix} \\
 &= \begin{pmatrix} a_{0,0}b_{0,0} & \cdots & a_{0,n-1}b_{0,m-1} \\ \vdots & \ddots & \vdots \\ a_{n-1,0}b_{m-1,0} & \cdots & a_{n-1,n-1}b_{m-1,m-1} \end{pmatrix}
 \end{aligned}$$

Partial Measurement

- If we use a certain method to measure the spin of both electrons in the $\mathbb{C}^4$ spaces, based on what we learned, it will randomly collapse to one of the basis states. For example,

$$|h\rangle_3 = \alpha_3|00\rangle + \beta_3|01\rangle + \gamma_3|10\rangle + \delta_3|11\rangle$$

If we measure $|h\rangle_3$, the probability it will collapse to $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$ is $|\alpha_3|^2$, $|\beta_3|^2$, $|\gamma_3|^2$, and $|\delta_3|^2$, respectively.

- E.g. to find the probability to collapse to $|10\rangle$ state, we can use the projector $P_{|10\rangle} = |10\rangle\langle 10|$ and $\langle h_3|P_{|10\rangle}|h_3\rangle$.

$$\begin{aligned}\langle h_3|P_{|10\rangle}|h_3\rangle &= (\alpha_3^*\langle 00| + \beta_3^*\langle 01| + \gamma_3^*\langle 10| + \delta_3^*\langle 11|)|10\rangle\langle 10|(\alpha_3|00\rangle + \beta_3|01\rangle + \gamma_3|10\rangle + \delta_3|11\rangle) \\ &= \gamma_3^*\langle 10||10\rangle\langle 10|\gamma_3|10\rangle \\ &= \gamma_3^*\gamma_3 = |\gamma_3|^2\end{aligned}$$

Here we have used the orthonormal properties of the basis vectors. We can say $|10\rangle$ is orthogonal to $|00\rangle$, $|01\rangle$, and $|11\rangle$ because they are different basis vectors. Or we can use the fact that $|10\rangle = |1\rangle \otimes |0\rangle$, then,

$$\begin{aligned}\langle 11||10\rangle &= (\langle 1|_1 \otimes \langle 1|_2)(|1\rangle_1 \otimes |0\rangle_2) \\ &= \langle 1|_1 |1\rangle_1 \times \langle 1|_2 |0\rangle_2 = 1 \times 0 = 0\end{aligned}$$

Partial Measurement

- We may also ask what is the probability to collapse to $|00\rangle$ or $|01\rangle$ states and we can use the projector $P_{|00\rangle|01\rangle} = |00\rangle\langle 00| + |01\rangle\langle 01|$ to find it. Similarly, we will get $|\alpha_3|^2 + |\beta_3|^2$.
- We may also only perform a measurement on one of the electrons, e.g. electron 1. The vector (state) for electron 1 can be written as $|f\rangle_1 = \alpha_1|0\rangle_1 + \beta_1|1\rangle_1$ and that for electron 2 as $|g\rangle_2 = \alpha_2|0\rangle_2 + \beta_2|1\rangle_2$.
- If after measurement, electron 1 collapses to $|0\rangle_1$, it will become $|f\rangle_1 = \alpha_1|0\rangle_1$ (not normalized yet) and only $|00\rangle$ and $|01\rangle$ will remain in the whole system. This is called the **partial measurement**.
- What is the probability that electron 1 will collapse to $|0\rangle_1$ in this partial measurement? Since after electron collapses to $|0\rangle_1$, only $|00\rangle$ and $|01\rangle$ are left in the *whole system*, so it should be the same asking “what is the probability to collapse to the $|00\rangle$ or $|01\rangle$ states” after the measurement? Therefore, it is $|\alpha_3|^2 + |\beta_3|^2$.
- Another way we can check if this true by using the projector operator, $P_{|0\rangle_1} = |0\rangle_1\langle 0|_1$ to confirm.
- The last way is that we can also see this as electron 1 firstly collapses to $|0\rangle$ and then performs tensor product with $|g\rangle_2$ to obtain the vector in the $\mathbb{C}^4$ space,

$$|h\rangle_3 = \alpha_1|0\rangle_1 \otimes (\alpha_2|0\rangle_2 + \beta_2|1\rangle_2)$$

$$= \alpha_1\alpha_2|0\rangle_1|0\rangle_2 + \alpha_1\beta_2|0\rangle_1|1\rangle_2$$

$$= \alpha_3|0\rangle_1|0\rangle_2 + \beta_3|0\rangle_1|1\rangle_2$$

However, We need to **normalize** the vector which is required.

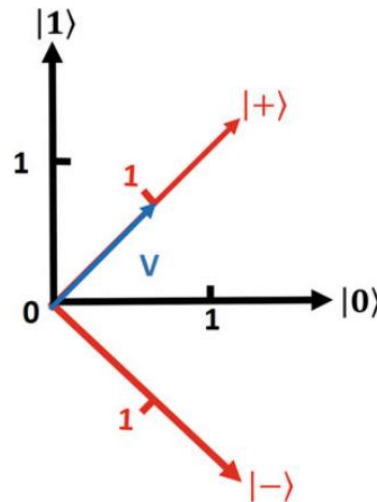
So, the probabilities of obtaining $|0\rangle_1|0\rangle_2$ and $|0\rangle_1|1\rangle_2$ sum up to 1.

Matrix Representation in a Given Basis

- We know that in different bases, a vector has different representations in column or row forms. For example,

$$|V\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- The above vector is in the $|0\rangle/|1\rangle$ basis (black) but $|V\rangle = |+\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ in the $|+\rangle/|-\rangle$ basis (red).



Pic.8 Representations of $|V\rangle$ in two different bases

Matrix Representation in a Given Basis

- Naturally, the operator should also have different representations in different bases. How do we represent an operator, $\mathbf{H}$, in a given basis? Assume the basis has basis vectors $|\alpha_k\rangle$ where k running from 0 to $n - 1$ in an n -dimensional space. The i -th row and j -th column elements is given by:

$$H_{i,j} = \langle \alpha_i | H | \alpha_j \rangle$$

Note that $H_{i,j}$ is a scalar (complex number). What is the representation of σ_x in the basis form by the eigenvectors of σ_z ?

- Firstly, the eigenvectors of σ_z from a 2D space. The two eigenvectors are $|\alpha_0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|\alpha_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. So the matrix must be 2×2 , with k running from 0 to 1. For example, $H_{0,0} = \langle \alpha_0 | \sigma_x | \alpha_0 \rangle$. But we need to know σ_x in order to evaluate this equation. So, it is a chicken and egg problem.
- Since $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, we have

$$\begin{aligned} \sigma_x &= \begin{pmatrix} \langle \alpha_0 | \sigma_x | \alpha_0 \rangle & \langle \alpha_0 | \sigma_x | \alpha_1 \rangle \\ \langle \alpha_1 | \sigma_x | \alpha_0 \rangle & \langle \alpha_1 | \sigma_x | \alpha_1 \rangle \end{pmatrix} \\ &= \begin{pmatrix} (1 \ 0) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} & (1 \ 0) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ (0 \ 1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} & (0 \ 1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{aligned}$$

Summary

- **Tensor Product Proficiency**
 - Gained familiarity with operations on both vectors and matrices.
- **Vector Decomposition**
 - Decomposing higher-dimensional vectors into lower-dimensional tensor products.
 - Note: This is not always possible (a precursor to Entanglement).
- **Partial Measurement**
 - Understanding the measurement of specific sub-space vectors within a larger system.
- **Matrix Construction**
 - Learned the procedure to construct a matrix representation from a given basis.